

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Developed and maintained **backend AI-powered microservices** using **Python, FastAPI, and Flask** integrated with **LangChain and Semantic Kernel frameworks** for multi-agent LLM orchestration, reducing model response latency by **40%** through prompt engineering and RAG pipeline optimization across 8+ enterprise deployments.
- Built end-to-end **GenAI application front-ends** using **React, Angular, and Vue.js** with **REST and GraphQL API integration** to backend services, implementing accessible UI components and standards-compliant design patterns; collaborated with UX designers and product managers to deliver **5 production-grade AI applications** serving **500+ concurrent users**.
- Applied **MLOps practices including model deployment, monitoring, and lifecycle management** via **Azure ML and AWS SageMaker**, containerizing models with **Docker** and orchestrating CI/CD pipelines using **Jenkins**; established comprehensive model documentation including strengths, limitations, and data-driven performance justifications reducing model drift incidents by **65%**.
- Designed and implemented **secure, compliant AI systems** adhering to **GDPR, EU AI Act, and Responsible AI guidelines**; conducted regulatory impact assessments and embedded audit trails across **3 banking-domain AI solutions**, ensuring data protection and ethical AI governance in production environments.
- Engineered **prompt engineering experimentation frameworks** supporting evaluation of **multiple LLM models (GPT-4, Claude, Llama)** with standardized testing methodologies; built A/B testing infrastructure for AI model performance comparison, improving model selection accuracy by **35%** and reducing deployment risk.
- Led cross-functional collaboration with data engineers, QA teams, and junior developers through knowledge-sharing sessions and mentoring on **Agile/Scrum practices**; implemented automated testing suites covering **unit, integration, and end-to-end scenarios** using **pytest and Jest frameworks**, achieving **92% code coverage** across AI service modules.

SKILLS

Backend AI & LLM Development – Python, FastAPI, Flask, LangChain, Semantic Kernel, multi-agent systems, prompt engineering, model orchestration, RAG pipelines.

Frontend Development – React, Angular, Vue.js, HTML, CSS, JavaScript, TypeScript, Figma, Storybook, accessible UI design, standards-compliant components.

Cloud ML & MLOps – Azure ML, AWS SageMaker, Docker, model deployment, monitoring, lifecycle management, experiment tracking, model registry.

API Integration & Backend Orchestration – REST, GraphQL, backend service architecture, system integration, API security, request/response optimization.

AI Security & Compliance – GDPR, EU AI Act, Responsible AI guidelines, audit trails, data protection, regulatory compliance, ethical AI governance.

DevOps & Testing – Jenkins CI/CD, Git, automated testing (pytest, Jest), unit testing, integration testing, end-to-end testing, code quality management.

Agile & Collaboration – Scrum, Agile methodologies, Application Lifecycle Management, Jira, Confluence, cross-functional team mentoring, knowledge sharing.

EDUCATION AND TRAINING

MASTER OF SCIENCE in Data Analytics – National College of
Ireland, Dublin

02/2026

BACHELOR OF ENGINEERING in Computer Science – SNS College
of Technology, Coimbatore, India

01/2021

PROJECTS

Multi-Agent Financial Advisory Engine with LLM Orchestration (HCL Technologies)

- Architected and deployed a **multi-agent AI system** using **Semantic Kernel and LangChain** frameworks orchestrating **GPT-4 and specialized LLMs** to generate real-time financial compliance advice and regulatory insights for banking clients; integrated **Azure ML** for model training and monitoring, reducing regulatory response time by **58%**.
- Engineered **React.js and TypeScript front-end interface** with **GraphQL API integration** to multi-agent backend services, implementing interactive prompt experimentation dashboards and model explanation UIs compliant with **WCAG 2.1 accessibility standards**; enabled business users to test **15+ prompt variants** and compare LLM outputs side-by-side.
- Embedded **GDPR and EU AI Act compliance controls** into the AI pipeline including data lineage tracking, consent audit trails, and model bias monitoring; documented model strengths, limitations, and data-driven performance justifications in comprehensive technical specifications reviewed by banking compliance and legal teams.
- Implemented **MLOps lifecycle management** with **Docker containerization, Jenkins CI/CD pipelines, and AWS SageMaker endpoints** enabling weekly model retraining and A/B testing across **8 LLM variants**; achieved **99.2% uptime** and reduced model deployment cycle time from **2 weeks to 3 days**.
- Delivered **pytest and Jest automated test suites** covering unit, integration, and end-to-end scenarios for prompt chains and API contract validation; mentored **4 junior engineers** on MLOps best practices and secure AI development patterns, ensuring **90%+ code coverage** across all deployments.

GenAI-Powered Payments Operations Dashboard with Prompt Engineering Framework (HCL Technologies)

- Built production **Python backend API** using **FastAPI** to expose **SWIFT MT/MX payment message processing** logic and **ISO 20022 compliance rules** to a **Vue.js GenAI front-end** powered by **LLM-driven natural-language query parsing**; supported **1,200+ daily transactions** with **99.8% uptime** and **sub-200ms response latency**.
- Designed and executed **prompt engineering experimentation framework** comparing **Claude, GPT-4, and specialized financial LLMs** for payment anomaly detection and SEPA compliance explanation; built rigorous evaluation metrics for accuracy, latency, and cost, selecting optimal models that reduced false-positive alerts by **47%**.
- Ensured **secure, compliant AI development** aligned with **GDPR, PSD2, and banking regulatory requirements**; implemented encrypted data flows for payment messages, role-based access control, and comprehensive audit logging for all LLM interactions with sensitive financial data.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California, Davis** (Coursera)

- Certificate of Completion in Python programming from **University of Michigan** (Coursera)